

Chunking: The Brain's Compression Algorithm

组块化：大脑的压缩算法

The gap between experts and novices is the size of their chunks · Printable Plate

I. THE COMPRESSION · DEMONSTRATION

Novice

JFKLHRCDGNRTPEKSYD

18 chunks

Expert

JFKLHRCDGNRTPESYD

6 chunks

The same 18 letters: a novice sees meaningless symbols, a frequent flyer sees 6 international airports (New York · London · Paris · Tokyo · Beijing · Sydney). Working memory capacity hasn't changed, but the **information density per chunk** differs by 3×.

II. OBSERVATIONS · OBSERVATIONS

Obs. 1

Chunks are "containers of meaning" — not arbitrary grouping, but structured compression based on prior knowledge. A phone number 138-0013-8000 is easier than 13800138000 because the former fits the "segment" pattern.

Obs. 2

Experts see the forest, novices see the trees — Chess masters remember 20+ piece positions not because of better memory, but because positions fit their "pattern library" (Chase & Simon, 1973). With random arrangements, masters and novices perform equally.

Obs. 3

Chunking is reversible — A chunk can be unpacked into sub-chunks or compressed into a larger chunk. Letters → words → phrases → conceptual frameworks; each level is a chunking of the previous.

Obs. 4

Automation is the endpoint of chunking — After thousands of repetitions, a chunk shifts from "conscious processing" to "automatic execution." This is why experienced drivers shift gears without thinking, while novices deliberate each step.

References

— Chase & Simon (1973). Perception in Chess. *Cognitive Psychology*, 4(1), 55-81. | Cowan (2001). The magical number 4 in short-term memory. *Behavioral and Brain Sciences*, 24(1), 87-114. | Miller (1956). The magical number seven, plus or minus two. *Psychological Review*, 63(2), 81-97.

Related Books

— *Why Don't Students Like School?* Daniel Willingham | *Peak: Secrets from the New Science of Expertise* Anders Ericsson

III. METHODS · PRACTICAL METHODS

1

Look for patterns, not details — When learning guitar, don't memorize "2nd string, 3rd fret"; memorize the "C chord shape." Once the pattern is established, details follow naturally.

2

Use outlines to layer information — A chunk needs 7 ± 2 connections to stabilize (Miller, 1956). Creating hierarchical structure (general → specific → example) compresses scattered information into portable chunks.

3

Mount new knowledge onto old chunks — Learning doesn't start from zero; it extends existing patterns. Teaching chess to a Go player is far faster than teaching a complete beginner.

4

Deliberate practice until automatic — After calling the same chunk thousands of times, it shifts from "occupying working memory" to "automatic execution." This is the only path from deliberate practice to fluent expression.

5

Teaching others is the best chunking test — If you can't explain a concept in simple terms, it hasn't been truly chunked. The Feynman Technique is essentially "compression verification."

* Core mechanism: Chunking reduces working memory load by compressing information, allowing limited cache to handle larger problems — this is the essence of expert thinking.

